

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Physics — Class IX

Syllabus: Ch 1 — Physical Quantities & Measurement | Ch 2 — Kinematics | Ch 3 — Dynamics-I
| Ch 4 — Dynamics-II | Ch 5 — Pressure & Deformation | Ch 6 — Work & Energy (till 6.4) | Ch
9 — Nature of Science

ANSWER KEY (Version 1)

Total Marks: 65

Time Allowed: 2 Hours 30 Minutes

Version: 1

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Answer	Explanation
1	B	From Newton's second law $F = ma$: units = $\text{kg} \times \text{m/s}^2$. One newton is the force that gives a 1 kg mass an acceleration of 1 m/s^2 , so $1 \text{ N} = 1 \text{ kg} \cdot \text{m/s}^2$.
2	B	A vernier caliper has a least count of 0.01 cm (0.1 mm). A metre rule has $\text{LC} = 1 \text{ mm}$; a screw gauge has $\text{LC} = 0.01 \text{ mm}$. The vernier's LC arises from the difference between the main scale division and the vernier scale division.
3	C	When a ball is thrown upward, gravity decelerates it (velocity decreases). Deceleration is negative acceleration because the acceleration due to gravity acts downward, opposite to the velocity direction.
4	B	Newton's first law (law of inertia) states that a body remains at rest or in uniform motion unless acted on by a net external force. It defines inertia as the natural tendency to resist change in state of motion.
5	B	Torque (moment of force) $\tau = F \times d$. Units = $\text{N} \times \text{m} = \text{N} \cdot \text{m}$. It is a vector quantity that produces rotational motion, analogous to force in translational motion.
6	B	Hooke's law holds only within the elastic limit (also called limit of proportionality). Beyond this point the spring permanently deforms and the force-extension graph curves away from the straight line.
7	C	Pascal's principle : pressure applied to an enclosed fluid increases equally at every point throughout the fluid. This is why a hydraulic lift can multiply force by having a larger output piston area.
8	C	Kinetic energy $E_k = \frac{1}{2} mv^2$. This formula is derived from the work-energy theorem. Note that KE depends on the square of speed, so doubling speed quadruples KE.
9	C	$E_p = mgh = 5 \text{ kg} \times 10 \text{ m/s}^2 \times 4 \text{ m} = \text{200 J}$. The work done against gravity in lifting the object is stored as gravitational potential energy.
10	C	Power $P = W/t$. The SI unit is the watt (W), named after James Watt. $1 \text{ W} = 1 \text{ J/s}$. For practical purposes a larger unit, the horsepower ($1 \text{ hp} = 746 \text{ W}$), is also used.
11	B	A scientific theory is a well-supported, rigorously tested explanation of natural phenomena. It is not a guess (that is a hypothesis) and it is more than just an observation; it has predictive and explanatory power.
12	D	Velocity is a vector quantity as it has both magnitude (speed) and direction. Speed, mass and temperature are scalar quantities with magnitude only. Head-to-tail method is used to add vectors.

SECTION B — Short Questions (11 × 3 = 33 Marks)

2(i) Base and derived quantities / Scalars and vectors 3 marks

Base quantities are independent fundamental quantities from which all others are derived. There are 7 SI base quantities. 1

Examples: *length* (metre, m) and *mass* (kilogram, kg).

Derived quantities are obtained by combining base quantities through multiplication or division. 1

Examples: *velocity* ($\text{m/s} = \text{m} \div \text{s}$) and *force* ($\text{N} = \text{kg} \cdot \text{m/s}^2$).

The table of SI base units is a key reference in all physics calculations. 1

OR

Scalar quantities have magnitude only (e.g., distance = 5 m, speed = 20 m/s). 1

Vector quantities have both magnitude and direction (e.g., displacement = 5 m North, force = 10 N downward). 1

Vectors are represented graphically by arrows: the length represents magnitude and the arrowhead shows direction. Two vectors are added using the **head-to-tail rule**. 1

2(ii) Distance and displacement / Speed and velocity 3 marks

Distance is the total path length covered (scalar). **Displacement** is the shortest distance from start to finish point in a specified direction (vector). 1

Walking 3 km North then 4 km East: total distance = $3 + 4 = 7 \text{ km}$. 1

Displacement = $\sqrt{(3^2 + 4^2)} = \sqrt{(9 + 16)} = \sqrt{25} = 5 \text{ km}$ (at an angle NE). 1

OR

Speed = distance/time (scalar). **Velocity** = displacement/time (vector, has direction). 1

Uniform velocity: equal displacement in equal time intervals (e.g., car on motorway at constant 100 km/h in one direction). **Variable velocity**: direction or magnitude changes (e.g., a car turning a corner). 1

The **gradient of a distance-time graph** gives the speed of the object. A steeper slope means greater speed. 1

2(iii) Newton's second law / Momentum and impulse 3 marks

Newton's second law: the acceleration of an object is directly proportional to the net force and inversely proportional to its mass: $F_{\text{net}} = ma$. 1

Acceleration = $F/m = 60 \text{ N} / 12 \text{ kg} = 5 \text{ m/s}^2$. 1

From $F = ma$, the SI unit of force is **newton (N)** = $\text{kg} \cdot \text{m/s}^2$. One newton accelerates a 1 kg mass at 1 m/s^2 . 1

OR

Momentum $p = mv$ (vector, SI unit: $\text{kg} \cdot \text{m/s}$ or $\text{N} \cdot \text{s}$). 1

Impulse $J = F \times \Delta t$ (force $\times$ time). From Newton's second law: $F = \Delta p / \Delta t$, therefore $F \times \Delta t = \Delta p$. 1

So **impulse = change in momentum**. Both have the same SI unit $\text{N} \cdot \text{s}$ (or $\text{kg} \cdot \text{m/s}$). Practical example: catching a ball slowly increases Δt , reducing the force felt. 1

2(iv) Newton's third law / Conservation of momentum 3 marks

Newton's third law: whenever object A exerts a force on object B, object B exerts an equal and opposite force on A. $F_{AB} = -F_{BA}$. 1

Example 1: When we jump, legs push the ground (action) and the ground pushes us upward (reaction). 1

Example 2: A rocket's exhaust gases push downward (action) while the gases push the rocket upward (reaction). Action-reaction pairs act on *different* objects. 1

OR

Law of conservation of momentum: the total momentum of an isolated system remains constant if no net external force acts on it. $m_1 v_1 = m_2 v_2$. 1

Initial momentum = 0 (both at rest). After firing: $m_b v_b + m_g v_g = 0$. 1

$v_g = -m_b v_b / m_g = -(0.03 \times 300) / 3 = -3 \text{ m/s}$ (recoil, opposite to bullet direction). 1

2(v) Torque and principle of moments / Equilibrium conditions 3 marks

Torque $\tau = F \times d$: the turning effect produced by a force about a fixed point, where d is the perpendicular distance (moment arm). SI unit: $\text{N}\cdot\text{m}$. 1

Principle of moments: for an object in equilibrium, the sum of clockwise moments equals the sum of anticlockwise moments about any pivot. 1

Torque = $40 \text{ N} \times 0.25 \text{ m} = 10 \text{ N}\cdot\text{m}$. 1

OR

Equilibrium: a state in which all individual forces and torques on an object are balanced (net force = 0 AND net torque = 0). 1

First condition: $\Sigma F = 0$ (no translational acceleration). **Second condition**: $\Sigma \tau = 0$ (no rotational acceleration). Both must hold for complete equilibrium. 1

Static equilibrium: object at rest (e.g., book on a table). **Dynamic equilibrium**: object moving at uniform velocity or uniform rotation (e.g., paratrooper falling at terminal velocity, ceiling fan at constant speed). 1

2(vi) Types of equilibrium / Friction 3 marks

Stable equilibrium: when displaced, the object returns to its original position; centre of mass rises on displacement (e.g., cone resting on its base). 1

Unstable equilibrium: when displaced, the object moves further away; centre of mass falls (e.g., pencil balanced on its tip). **Neutral equilibrium**: when displaced, object stays in its new position; CM neither rises nor falls (e.g., ball on flat surface). 1

A **lower centre of gravity** and wider base improve stability. Racing cars and buses are designed with low CG to resist toppling. 1

OR

Friction: a contact force that resists relative motion between surfaces in contact; acts opposite to motion. SI unit: newton (N). Types: **sliding**, **rolling** (much smaller), **fluid friction** (drag). 1

Advantages: enables walking (friction between shoes and ground); car brakes work through friction. 1

Disadvantages: causes wear and tear of machine parts; wastes energy as heat in engines. Rolling friction is much less than sliding friction, which is why wheels replaced sledges. 1

2(vii) Hooke's law and spring constant / Pressure 3 marks

Hooke's law: within the elastic limit, the extension x of a spring is directly proportional to the applied force F . Mathematically: $F = kx$, where k is the **spring constant**. 1

Spring constant k = restoring force per unit extension. SI unit: N/m . 1

$k = F/x = 20 \text{ N} / 0.05 \text{ m} = 400 \text{ N/m}$. 1

OR

Pressure $P = F/A$ (force per unit area). SI unit: pascal (Pa) = N/m^2 . 1

(a) Force constant, area decreases: pressure increases. Example: a sharp knife blade has a tiny area, so the same hand force creates very high pressure to cut food. 1

(b) Area constant, force increases: pressure increases. Example: pressing a drawing pin harder increases the force, increasing pressure at the tip and making it penetrate deeper. 1

2(viii) Atmospheric pressure and barometer / Liquid pressure derivation 3 marks

Atmospheric pressure: the pressure exerted by the weight of the air column above a surface. At sea level it equals approximately $1.013 \times 10^5 \text{ Pa}$ (1 atm = 101.325 kPa). 1

A **mercury barometer** consists of a mercury-filled tube inverted into an open mercury trough. Atmospheric pressure pushes mercury up the tube until the weight of the mercury column balances the atmospheric pressure. At sea level the column rises to about **760 mm of Hg**. 1

Atmospheric pressure decreases with altitude because there is less air above; higher altitudes have lower pressure. This is why water boils at lower temperatures on mountains. 1

OR

Consider a liquid column of height h , density ρ , base area A . Mass of liquid $m = \rho Ah$; weight $F = \rho Ahg$. 1
Pressure at base $P = F/A = \rho Ahg/A = P = \rho gh$. 1

Liquid pressure increases with depth because the weight of liquid above increases. It does not depend on the shape of the container (hydrostatic paradox). 1

2(ix) Work done / Conservation of energy 3 marks

Work done $W = F \times d$ (force times displacement in direction of force). SI unit: joule ($J = N \cdot m$). 1

$W = 150 \text{ N} \times 5 \text{ m} = \mathbf{750 \text{ J}}$. 1

Work done is zero when force is perpendicular to displacement (e.g., a satellite in circular orbit: centripetal force is perpendicular to velocity, so no work is done on the satellite). 1

OR

Law of conservation of energy: energy cannot be created or destroyed; it can only be converted from one form to another. Total energy of an isolated system remains constant. 1

(a) PE at top = $mgh = 2 \times 10 \times 10 = \mathbf{200 \text{ J}}$. 1

(b) By conservation of energy, KE just before ground = PE at top = **200 J** (all PE converted to KE, since air resistance neglected). 1

2(x) Power and efficiency / Renewable vs non-renewable energy 3 marks

Power $P = W/t$ (rate of doing work). SI unit: watt (W). **Efficiency** $\eta = (\text{useful output energy} / \text{input energy}) \times 100\%$. 1

(a) Efficiency = $(4800/6000) \times 100\% = \mathbf{80\%}$. 1

(b) Useful power output = $4800 \text{ J} / 20 \text{ s} = \mathbf{240 \text{ W}}$. 1

OR

Non-renewable resources: finite, cannot be quickly replaced; e.g., coal, petroleum/oil, natural gas, nuclear fuel (uranium). 1

Renewable resources: naturally replenished; e.g., solar radiation, wind energy, hydroelectric power, biomass, tidal energy. 1

Renewable sources are preferred because they do not emit greenhouse gases, will not run out, and reduce dependence on imported fuels. However, some (like solar and wind) are intermittent. 1

2(xi) Hypothesis, theory and law / Science, technology and engineering 3 marks

Hypothesis: a tentative assumption or explanation made before any research, which can be tested and verified (e.g., “heavier objects fall faster than lighter ones” before Galileo’s experiments). 1

Theory: a well-supported description of natural phenomena verified through rigorous testing; explains how nature behaves (e.g., *Kinetic Molecular Theory*, *Theory of General Relativity*). 1

Law: a statement of an observed regularity giving the relationship between variables; describes what happens but not always why (e.g., *Newton’s Law of Universal Gravitation*, *Newton’s Laws of Motion*). 1

OR

Science: pursuit of knowledge through experimentation and observation to understand how nature works (e.g., discovering Newton’s laws of motion). 1

Technology: application of scientific knowledge to create tools, machines and systems for practical purposes (e.g., designing a car engine using combustion principles). 1

Engineering: practical use of scientific and mathematical principles to design and construct structures and processes (e.g., civil engineers building bridges using structural mechanics). Science, technology and engineering are interconnected: science drives technology, and engineers use both. 1

SECTION C — Long Questions (4 × 5 = 20 Marks)

3(i) Acceleration and graphical derivation of equations of motion 5 marks

Acceleration: the rate of change of velocity. $a = \Delta v / \Delta t$. It is a vector quantity; SI unit m/s^2 . 1

Speed-time graph for uniform acceleration: a straight line from initial velocity v_i at $t = 0$ to final velocity v_f at time t . 1

v (m/s) v_f | / | / | v_i | _____ / | | / | | _____ | _____ t (s) θ t Gradient = $(v_f - v_i)/t$ = acceleration Area under graph = distance travelled

First equation: from gradient: $a = (v_f - v_i)/t \Rightarrow v_f = v_i + at$ 1

Second equation: distance = area under graph = area of trapezoid = $\frac{1}{2}(v_i + v_f) \times t$. Substituting v_f : $d = v_i t + \frac{1}{2}at^2$ 1

Two key readings from a speed-time graph: (1) the **gradient** gives acceleration; (2) the **area under the curve** gives the distance (displacement) covered. 1

OR

Translatory motion: every part of the body moves in the same direction. Types: *rectilinear* (straight line, e.g., a car on a straight road), *curvilinear* (curved path, e.g., projectile), *random* (irregular, e.g., gas molecules). 1

Rotatory motion: body spins about an axis (e.g., Earth rotating on its axis, fan blades). **Vibratory motion:** oscillation about a fixed point (e.g., pendulum, guitar string). 1

Distance: total path covered (scalar). **Displacement:** shortest path from start to end with direction (vector). 1

Speed: rate of change of distance (scalar, $v = s/t$). **Velocity:** rate of change of displacement (vector, $v = d/t$, includes direction). 1

Examples: rectilinear — train on straight track; rotatory — spinning top; vibratory — clock pendulum. 1

3(ii) Newton's three laws / Force and rate of change of momentum 5 marks

First Law (Inertia): if the net external force on an object is zero, it will remain at rest or continue in uniform straight-line motion. $F_{\text{net}} = 0 \Rightarrow \Delta v = 0$. Example: passengers jerk forward when a bus suddenly brakes (tendency to stay in motion). 1

Second Law: $F_{\text{net}} = ma$. The acceleration is directly proportional to net force and inversely proportional to mass. Example: the same engine accelerates a car faster than a bus (more mass = less acceleration). 1

Third Law: $F_{AB} = -F_{BA}$. For every action there is an equal and opposite reaction acting on a different body. Example: when we walk, our foot pushes the ground backward (action) and the ground pushes us forward (reaction). 1

Limitations: Newton's laws fail at the atomic/subatomic scale (quantum mechanics applies) and at speeds approaching the speed of light (relativistic mechanics applies). 1

Newton's laws are a fundamental framework for classical mechanics; Albert Einstein's theory of relativity extends them for extreme speeds. 1

OR

Momentum $p = mv$ (vector; SI unit $\text{kg}\cdot\text{m/s}$). It represents the quantity of motion of a body. 1

From Newton's second law: $F = ma = m(v_f - v_i)/\Delta t = (mv_f - mv_i)/\Delta t = \Delta p / \Delta t$. The net force equals the rate of change of momentum. 1

For the car: initial momentum = $1200 \times 8 = 9600 \text{ kg}\cdot\text{m/s}$; final momentum = 0. 1

$\Delta p = 0 - 9600 = -9600 \text{ kg}\cdot\text{m/s}$. $F = \Delta p / \Delta t = -9600 / 5 = -1920 \text{ N}$. 1

The negative sign indicates the braking force acts opposite to the direction of motion. A larger mass or higher initial speed requires greater braking force. 1

3(iii) Pascal's principle and hydraulic lift / Elasticity and Hooke's law 5 marks

Pascal's principle: an external pressure applied to an enclosed fluid is transmitted unchanged to every point within the fluid. 1

Hydraulic lift working principle: two pistons of different cross-sectional areas (A_1 small, A_2 large) are connected by hydraulic fluid. A small force F_1 on piston 1 creates pressure $P = F_1/A_1$. This same pressure acts on piston 2: $F_2 = P \times A_2 = (A_2/A_1) \times F_1$. The force is multiplied by the ratio of areas. 1

[Small piston F_1 , area A_1] ---fluid--- [Large piston F_2 , area A_2] $P_1 = F_1/A_1$ Pascal's principle: $P_1 = P_2$ $F_2 = (A_2/A_1) \times F_1$ -- force multiplier

Weight of car $F_2 = 17,640 \text{ N}$. From Pascal's principle: $F_1 = (A_1/A_2) \times F_2 = (0.002/0.9) \times 17,640$ 1

$F_1 = 0.002222 \times 17,640 = 39.2 \text{ N}$. A small force of only 39.2 N lifts the 17,640 N car. 1

Applications of Pascal's principle: hydraulic car brakes, hydraulic jack, dentist's chair. The principle shows pressure acts perpendicular to surfaces in a fluid. 1

OR

Elasticity: the ability of a deformed body to return to its original shape and size when the deforming force is removed. Materials that do not recover are called inelastic (e.g., clay, plasticine). 1

Hooke's law: within the elastic limit, the extension x is directly proportional to the applied force F . $F = kx$, where k = spring constant (N/m). 1

Force-extension graph: straight line through origin up to point A (**limit of proportionality**); point B is the **elastic limit** (beyond which permanent deformation occurs); point C is **ultimate stress**; point D is the **breaking point**. The gradient of the straight portion = spring constant k . 1

Three practical applications: (1) **spring balance** (weighing scale uses extension to measure weight); (2) **balance wheel in mechanical watches** (hairspring returns to centre position); (3) **galvanometer** (hair spring makes deflection proportional to current). 2

3(iv) KE, PE and conservation of energy / Power, efficiency and 100% limit 5 marks

Kinetic energy: energy possessed by a body due to its motion. Derivation: a constant force F acts on mass m over displacement d starting from rest. Work done $= Fd$. From Newton's 2nd law: $F = ma$; from kinematics $v^2 = 2ad$, so $d = v^2/2a$. Therefore $W = ma \times v^2/2a = E_k = \frac{1}{2}mv^2$. 1

Gravitational potential energy: energy stored in a body by virtue of its height in a gravitational field. To lift mass m through height h , work $W = F \times h = mg \times h = E_p = mgh$. 1

Law of conservation of energy: energy can neither be created nor destroyed; it can only change form. Total energy $=$ constant: $E_k + E_p =$ constant. 1

Falling ball example: at height h , $E_p = mgh$, $E_k = 0$ (at rest). At height $h/2$, some PE converts to KE: $E_p = mgh/2$, $E_k = mgh/2$. At ground, $E_p = 0$, $E_k = mgh$. At every point: $E_k + E_p = mgh =$ constant. 1

This confirms mechanical energy is conserved throughout the fall (neglecting air resistance). 1

OR

Power $P = W/t = E/t$. SI unit: watt ($W = J/s$). It measures how quickly energy is converted. **Efficiency** $\eta =$ (useful energy output / total energy input). 1

(a) $\eta = (E_{out}/E_{in}) \times 100\% = (120/400) \times 100\% = 30\%$. 1

(b) Useful power $= E_{out}/t = 120 J / 0.5 s = 240 W$. 1

A machine can never be 100% efficient because frictional forces always convert some input energy into waste thermal energy. This waste heat is not useful for the intended purpose of the machine. 1

The law of conservation of energy is not violated; energy input always equals useful output plus waste heat ($E_{in} = E_{out} + E_{waste}$). Incandescent bulbs are only 5% efficient; fluorescent lamps about 20% efficient. 1